

Response to Reviewer 2

Automated Segmentation and Classification of Coral Reef Imagery Using Deep Learning

September 2026

We thank the reviewer for a thorough and constructive read. Below we address each comment in turn. Our responses appear in **red**; all section, table, and figure numbers refer to the revised manuscript (`submission/main.tex`).

Comment 1 — Novelty framing

Novelty framing overstated (“first” semantic segmentation approach for colony-scale bleaching assessment) given CoralSCOP and related literature. Clarify that the novelty is in the combination of SAM2 + bleaching status + morpho-taxonomic grouping + ASV top-down imagery, not a broad first-in-field claim.

Response: *The specific novelty claim has been narrowed* The Related Work paragraph (Section 2) was rewritten to situate the contribution relative to CoralSCOP Zheng et al. (2024) and other recent work. The original broad “first semantic segmentation pipeline for coral reef imagery” framing has been removed. The claim is now scoped specifically to the combination of features not present in any single prior system: SAM2-based mask proposal tuned for multi-scale coral detection, an ensemble CNN classifier for morpho-taxonomic class *and* bleaching status simultaneously, and application to top-down ASV imagery of a bleaching event. The Related Work closing sentence now reads: “Here we present (1) a novel pipeline combining SAM2 mask proposals with an ensemble CNN classifier for both morpho-taxonomic class and bleaching status, and (2) the first application of semantic segmentation to automate colony-scale bleaching classification from top-down ASV imagery” — narrowing the novelty claim to the specific combination not addressed by prior work.

Comment 2 — Data / annotation methodology

Data section: annotations were SAM2-assisted, which may bias evaluation (since the pipeline itself relies on SAM2). Report inter-annotator agreement, QC procedures, correction rules, and any independent expert audit. Clarify the “non-coral” class definition (negative masks weren’t explicitly annotated; they become a major training component via synthetic generation).

Response: *Partially addressed.* The revised manuscript adds an explicit description of the synthetic non-coral generation procedure (Section 3.2.3): a candidate SAM2 mask is assigned a non-coral label whenever its pixel overlap with any annotated coral polygon is at or below a fixed tolerance (`TOLERANCE` = 0.2), i.e. at most 20% of the candidate mask’s pixels may be genuine coral. A Limitations paragraph (Section 5) now names this as a source of label noise and frames the 0.2 threshold explicitly as a bias–variance tradeoff.

Remaining gaps:

- Inter-annotator agreement statistics and a description of the expert correction workflow have not been added. We acknowledge this is a meaningful gap: SAM2-assisted annotation could in principle inflate apparent segmentation quality if the model’s own mask boundaries inform the ground truth. A short methods paragraph on QC/correction procedures and, where available, an inter-annotator agreement metric (e.g. Cohen’s κ or mask-level IoU between annotators) should be added before final submission.
- An independent expert audit of a randomly sampled subset of annotations has not been conducted or reported.

Comment 3 — Experimental design / baselines / ablations

Experimental design under-validated: single reef system, single ASV platform, one camera geometry, held-out test set from the same domain. Add baseline comparisons (SAM2 alone, CoralSCOP, U-Net/DeepLab, Mask R-CNN, non-ensemble ResNet) and ablations (dual SAM2 config, synthetic non-coral generation, oversampling, augmentation, ensemble weighting, mask consolidation).

Response: *Partially addressed.* Two baselines have been added to the revised manuscript:

- **CoralSCOP** (Zheng et al., 2024): run on the same 552 images and scored with the identical pixel-level IoU/Dice definitions (Section 4.1, Table 7). NeuralReefer outperforms CoralSCOP on every metric in both splits; the comparison also includes an IoU exceedance curve (Figure 14).
- **Single CNN** (“Single CNN” in Table 7): the 5-model ensemble replaced with submodel 4 alone (otherwise identical pipeline), providing a non-ensemble ResNet

baseline. This isolates the ensemble’s contribution from the SAM2 mask proposal and consolidation stages.

Remaining gaps: SAM2-alone (no CNN classifier), U-Net/DeepLab, Mask R-CNN, and the full suite of ablations (dual SAM2 config, synthetic non-coral generation strategy, augmentation, oversampling, mask consolidation hyperparameters) have not been added. We acknowledge these would substantially strengthen the experimental validation, noting that each ablation requires a full retraining run.

Comment 4 — Train/validation/test terminology and leakage

Train/validation/held-out/test terminology is confusing; possible over-optimistic reporting. Abstract’s 90.9%/95.8%/94.3% appear to be from Table 4’s augmented validation set (1,762 masks), while ecological evaluation uses the 111 held-out images. Explicitly state that no masks, images, augmentations, or hyperparameters from the final 111-image test set were used anywhere upstream (SAM2 tuning, CNN training, ensemble weighting, threshold selection).

Response: Fully addressed. This comment received the most targeted set of changes in the revision:

1. **Abstract rewritten.** The abstract now leads with pipeline-level live coral cover IoU (0.674) and Dice (0.781) on the held-out test set as the primary performance claim, followed by a comparison against CoralSCOP (IoU 0.487/Dice 0.614). The mask-level classification figures (90.3% accuracy, 94.6% recall, 93.0% precision) are reported as a clearly labeled secondary result—“at the mask level, *once a candidate mask has been proposed*, the classification ensemble attains...”—so the reader cannot conflate the two quantities.
2. **Explicit no-leakage statement.** The data-partitioning paragraph in Section 3.1 now includes: “The 111 held-out test images were withheld entirely from all upstream steps: SAM2 configuration search, CNN training, ensemble weight estimation, and threshold selection were each conducted on the 441 training images and a separate mask-level validation set drawn from the same 441 images only.” A stacked-generalization justification (Wolpert 1992) is cited for the disjoint submodel/ensemble/test-set partitioning.
3. **Table 4 scope clarified.** The Table 4 note now reads: “All metrics in this table are mask-level classification statistics; pixel-level segmentation quality is reported separately in Table 6.” The body of Section 4.1 makes the causal point explicit: a classification ensemble that is accurate on proposed masks (Table 4) cannot recover coral that SAM2 never proposed in the first place, so pipeline-level IoU/Dice is treated as the primary end-to-end quality measure.

Comment 5 — Method description ambiguities

Method description has logical ambiguities:

- *Section 3.2.1’s mask consolidation (uses classification confidence) creates an apparent ordering conflict with Figure 4 (implies classify $\rightarrow$ reject non-coral $\rightarrow$ consolidate) vs. Sections 3.2.1/3.3 (implies consolidate $\rightarrow$ classify). Needs an unambiguous rewrite.*
- *Ensemble aggregation: non-coral class weight not reported.*
- *Optimization details incomplete.*
- *Convexity claim needs justification.*

Response: Fully addressed across multiple passes.

1. **Pipeline ordering.** Figure 4 was already correct. The prose in Sections 3.2.1 and 3.3 (“Coral Cover Estimation”) had incorrectly described the reverse order and has been rewritten to match the code and figure, with an explicit note that the subsections’ narrative grouping (segmentation $\rightarrow$ classification) differs from the execution order.
2. **Non-coral weight.** $\nu_{\text{noncoral}} = 0.25$ is now reported and justified in Section 3.2.4: the value matches an *a priori* prior of roughly 25% coral cover across the survey, boosting recall for the minority coral classes. A precise argument (citing Lipton, Elkan & Narayanaswamy 2014 on F_β -optimal Bayes classifiers) is provided showing that $\nu_{\text{noncoral}} = 1/4$ is consistent with F_2 -optimal classification ($\beta^2 = 4$ recall-weighting).
3. **Optimization details.** A full Expectation-Maximization / Minorize-Maximize derivation for the ensemble weight updates has been added as Appendix C, including the complete E-step, M-step, closed-form multiplicative update for $\mathbf{W}$, and renormalization step. A summary E-step/M-step description with an MM-surrogate interpretation remains in the main text (Section 3.2.5); the full six-equation derivation is deferred to the appendix to preserve readability.
4. **Convexity claim.** The original “novel convex ensemble optimization scheme” framing has been removed. Section 3.2.5 now explicitly states that the joint $(\boldsymbol{\alpha}, \mathbf{W})$ optimization is non-convex; only the individual conditional updates (holding one set of parameters fixed) are convex. The contribution is reframed around the interpretability of the learned responsibility decomposition.

Comment 6 — Conservative interpretation of results; IoU/Dice/CIs

Results should be interpreted more conservatively: pixel accuracy $>90\%$ can be inflated by dominant background/easy non-coral pixels. Add IoU, Dice, F1, per-class P/R, macro-averaged metrics, confidence intervals. Treat LCC/BCC underestimation as a substantive

limitation, not a minor residual bias.

Response: Substantially addressed.

1. **New Section 4.1 (“Segmentation Quality: IoU and Dice”).** This section now serves as the primary results section for end-to-end pipeline quality, explicitly noting that pixel accuracy can be inflated by the dominant non-coral background and that IoU/Dice are the appropriate summary statistics.
2. **Table 6 (per-taxonomy pixel IoU/Dice/precision/recall).** Reports pixel-level IoU, Dice, precision, and recall for all 10 taxonomic classes (LCC, bleached, and 8 genera), macro-averaged over the 111 test images. Standard errors ($SD/\sqrt{n}$, where $n = 111$ images) are reported for every cell in a separate column.
3. **Table 7 + Figure 14 (CoralSCOP baseline and IoU exceedance curve).** Both NeuralReefer and the Single CNN ablation are compared against CoralSCOP on live coral cover IoU/Dice/precision/recall, in-sample and out-of-sample, with standard errors throughout (Table 7). An IoU exceedance curve (Figure 14) shows the full distribution of per-image IoU for both methods: 81.1% of NeuralReefer’s test predictions reach $\text{IoU} \geq 0.5$ vs. 50.5% for CoralSCOP.
4. **LCC/BCC underestimation reframed.** The bias decomposition discussion (Table 5, Section 4) now treats the observed underestimation as a substantive limitation driven by the segmentation bottleneck. Section 4.1 makes the mechanism explicit: SAM2’s failure to propose a mask over a colony propagates into every downstream metric regardless of classification accuracy.
5. **Standard errors vs. bootstrap CIs.** Standard errors ($SD/\sqrt{n}$) are reported in Tables 6, 7, and A.8. The reviewer’s original request was for confidence intervals, which can be constructed from the SEs. If the reviewer prefers bootstrap CIs specifically, we are happy to add them.

Comment 7 — Figure quality

Figure quality: Figures 4, 5 too dense/small text. Figure 13 panels hard to read. Figure 15 negative depth values need clarification. Figure 1 needs scale/depth/quality context. Figure 9 needs clearer legends. Table 1 should clarify whether “average coral cover” includes both healthy+bleached; Table 4 should define whether P/R are coral-vs-noncoral, macro-averaged, or mask-level aggregated.

Response: Substantially addressed.

1. **Figures 4 and 5.** Both converted to full-page landscape (`sidewaysfigure`) on their own dedicated pages, substantially increasing text legibility. All figures in the paper now share a unified theme with a serif Computer Modern font, 40–60pt base text, and a consistent panel background.

2. **Figure 13.** The 12-panel density-plot figure was removed and replaced by Table 6 (per-class IoU/Dice), which conveys the same per-taxonomy information more precisely and compactly.
3. **Figure 15 / depth sign convention.** A new Appendix B (“Segmentation Accuracy vs. Water Depth”) explains the sign convention explicitly: the x-axis is depth *below the water surface* (0 = surface, negative = deeper), not distance to the reef substrate. The figure has been converted to a two-panel design (Full Range + Zoomed to 0–7.5m depth and 75–100% accuracy), and the caption and appendix prose both state the convention clearly.
4. **Figure 1.** The caption now states the pixel dimensions of each panel (1024×1024 px) and the survey’s camera depth-below- surface range (1.6–18.7 m). Physical scale bars and ground sample distance cannot be reported: Yellowfin’s metadata records depth below the water surface, not altitude above the reef substrate, so no camera-to-substrate distance is available to compute a GSD. This limitation is noted explicitly in the caption. Additionally, Figure 1 was rebuilt as a true scatter-of-images plot to show spatial and temporal context: panels now sit at their literal acquisition time (x-axis) and logged camera depth (y-axis) rather than a fixed 3×3 grid, giving the reader spatial and temporal context for the nine examples.
5. **Figure 9 legends.** shared color legend was added beneath each row.
6. **Table 4 metric definitions.** The table note now reads: “All metrics are mask-level classification statistics: a mask is a true positive if it is predicted as coral (any class), regardless of genus. Precision and recall are therefore coral-vs-noncoral, not macro-averaged across genera.” Table 4 is explicitly scoped to mask-level classification; pixel segmentation quality is in Table 6.
7. **Table 1 coral cover definition.** LCC is defined in the paper to include both healthy and bleached coral (Section 3.1).

Comment 8 — Related Work citations

Related Work lacks recent underwater/UAV monitoring literature: cites specific papers to add (Holothurians monitoring UAV 2025; Mussel-YOLO EI 2025; RBL-YOLO MPB 2024; multi-label coral condition monitoring MPB 2026; multi-scale coral condition/Drupella MPB 2026; Cost-Effective AUAV monitoring MER 2026).

Response: *Not yet addressed.* The six papers named by the reviewer have not been added to the Related Work section.

Comment 9 — Copyediting

Numerous typos (“heatstress,” “stake holders,” “over fishing,” “land based,” “long term,” “limited shallower reefs,” “photodata,” and others) and an overly informal acknowledgement. Also: “Beijorn” should be “Beijbom,” “Marcohorn” should be “Macrohon,” SAM2 is cited with the original SAM reference in places, and CoralMask/USIS10K/CoralSCOP references appear mismatched or duplicated.

Response: Substantially addressed. All individually named typos have been corrected in the revised manuscript:

- “heatstress” → “heat stress”
- “stake holders” → “stakeholders”
- “over fishing” → “overfishing”
- “land based” → “land-based”
- “long term” → “long-term”
- “limited shallower reefs” → “limited to shallower reefs”
- “photodata” → “photo data”
- “yielding to the complexity,” “proceeding section,” “helps balances,” “use to model,” “the proposed pipelines ability,” “proposed pipelines reproduces,” “may reflects” — all corrected
- “Beijorn” → “Beijbom”
- “Marcohorn” → “Macrohon”
- Informal acknowledgement rewritten: “A massive thank you goes [to] Aidan Quigley” → “We thank Aidan Quigley for helping to initiate this collaboration.”
- SAM2 now correctly cites Ravi et al. (2024) (not the original SAM, Kirillov et al. 2023) at its first reference.

References

Zheng, Z., Liang, H., Hua, B.-S., Wong, Y. H., Ang, P., Chui, A. P. Y., and Yeung, S.-K. (2024). Coralscop: Segment any coral image on this planet. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 28170–28180.